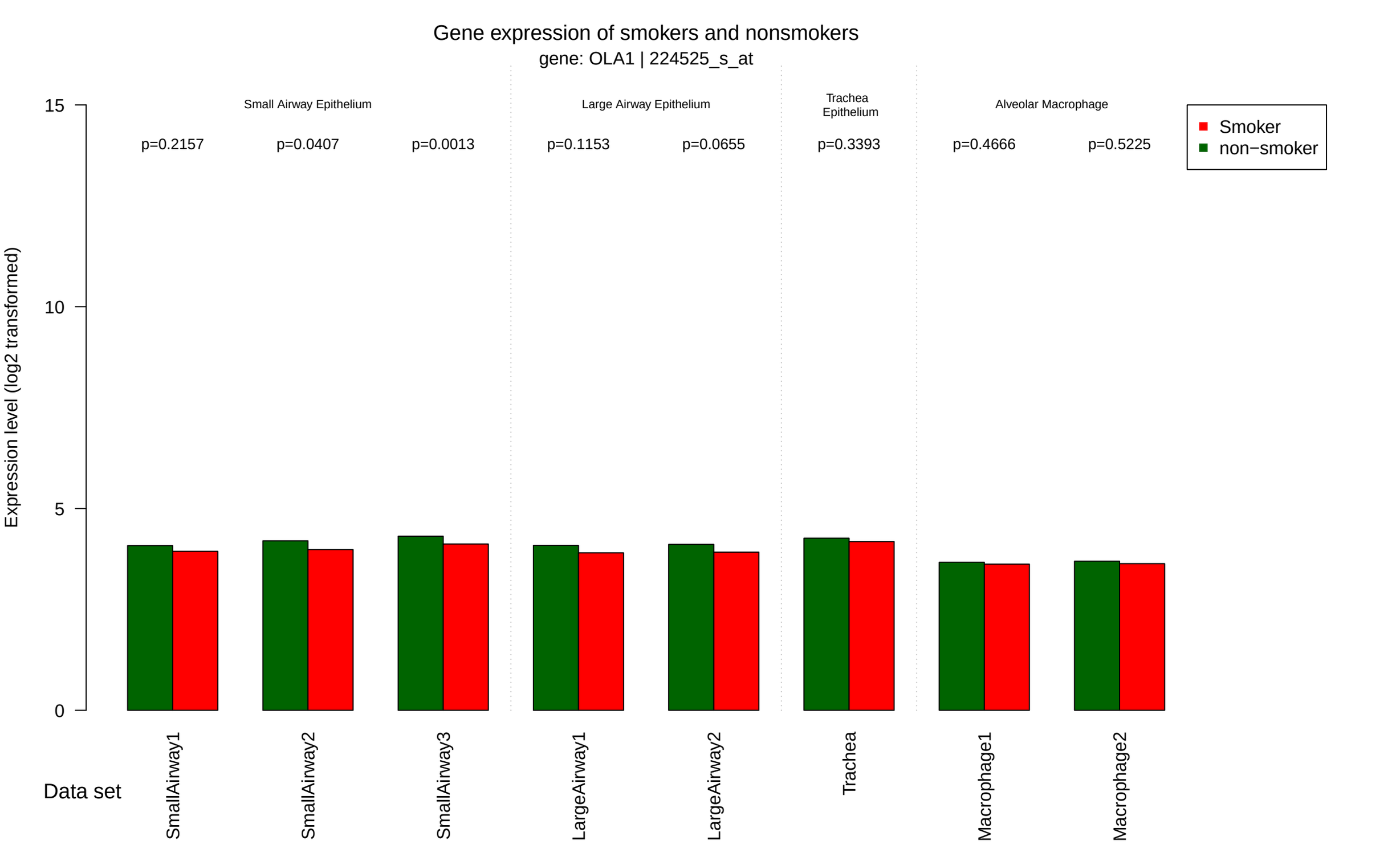
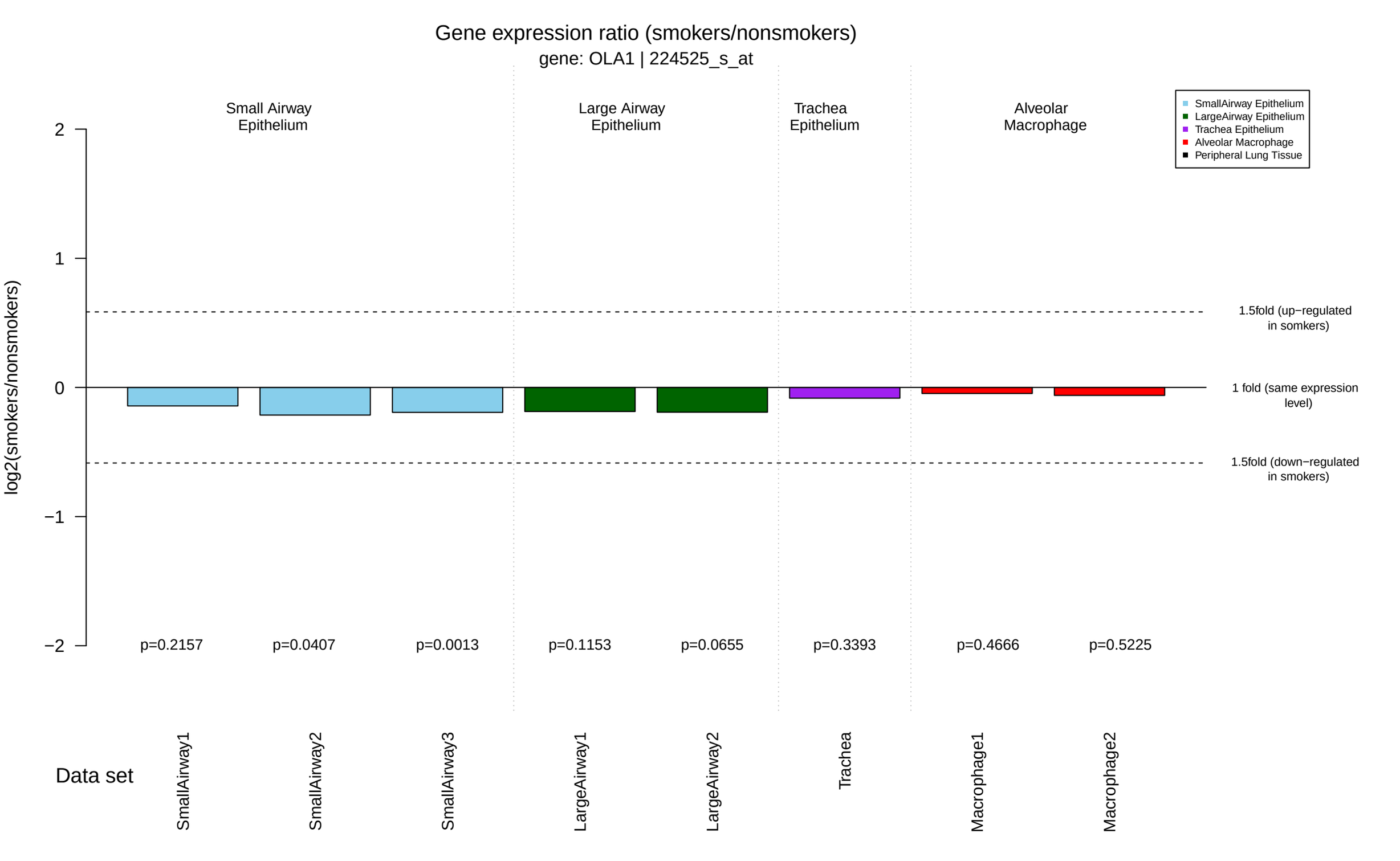
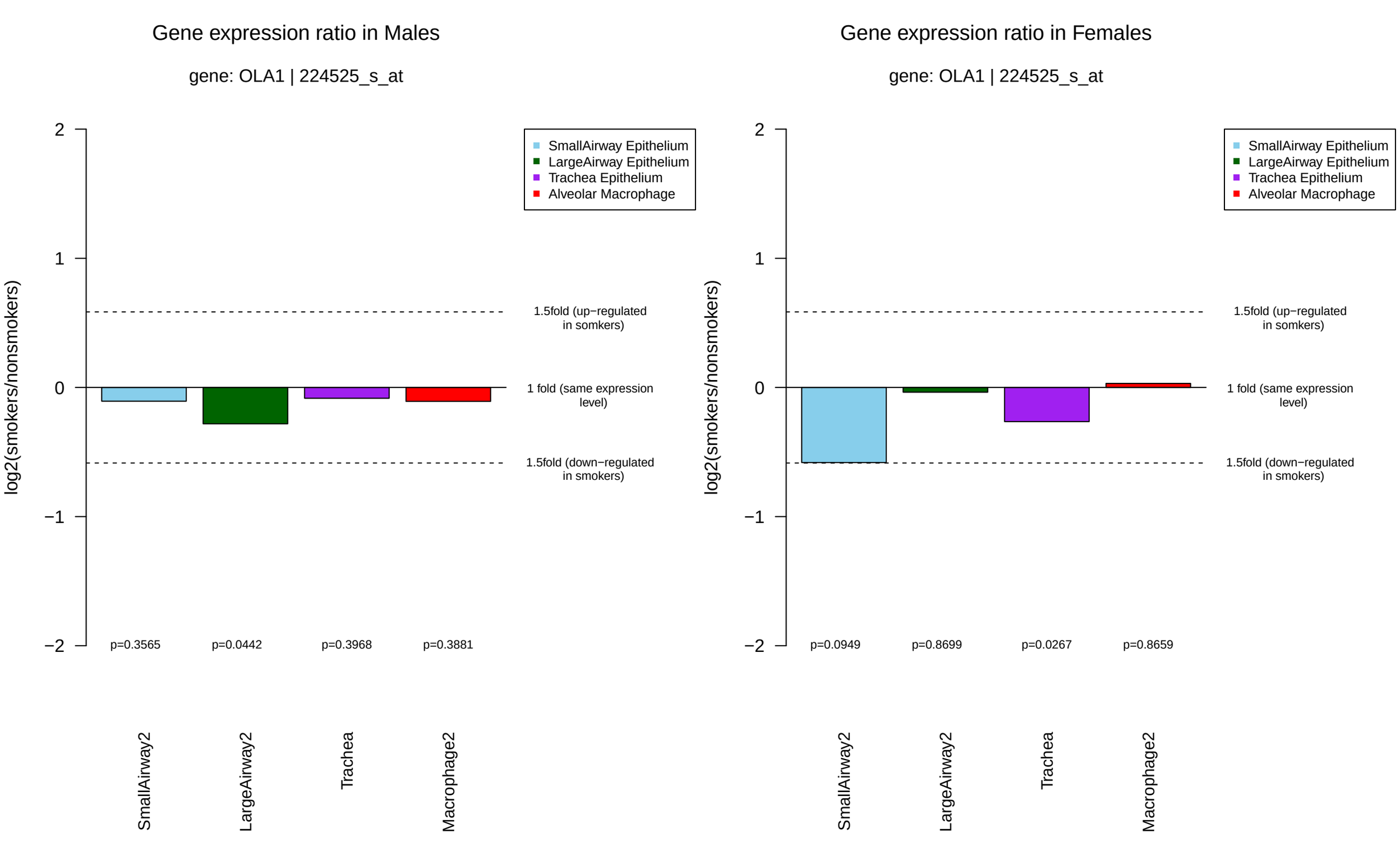
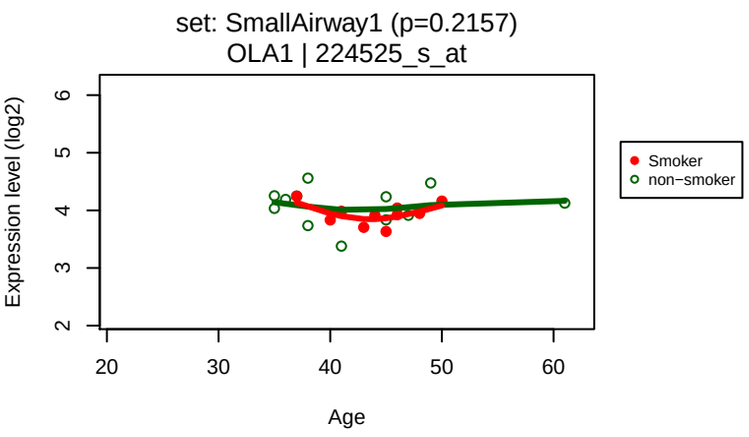
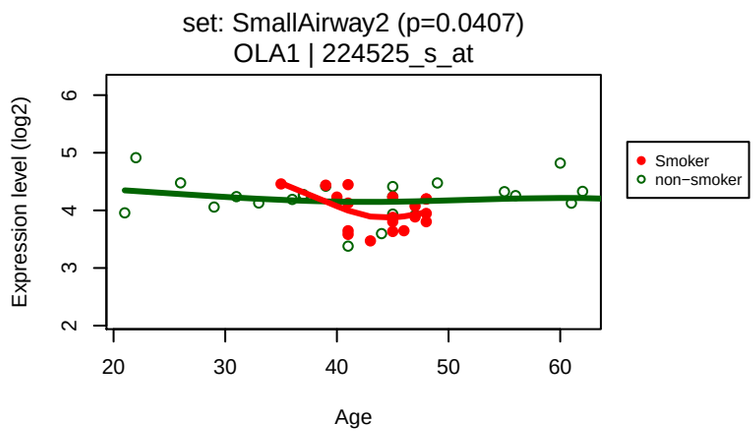
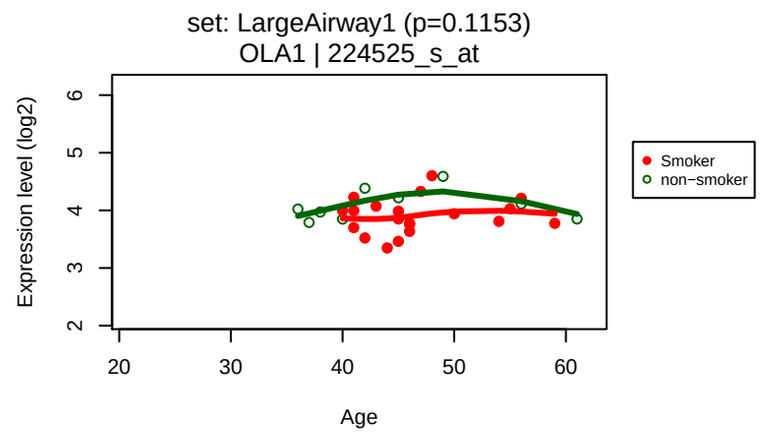

Gene expression of smokers and nonsmokers
gene: OLA1 | 224525_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2157
p=0.0407
p=0.0013
p=0.1153
p=0.0655
p=0.3393
p=0.4666
p=0.5225
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

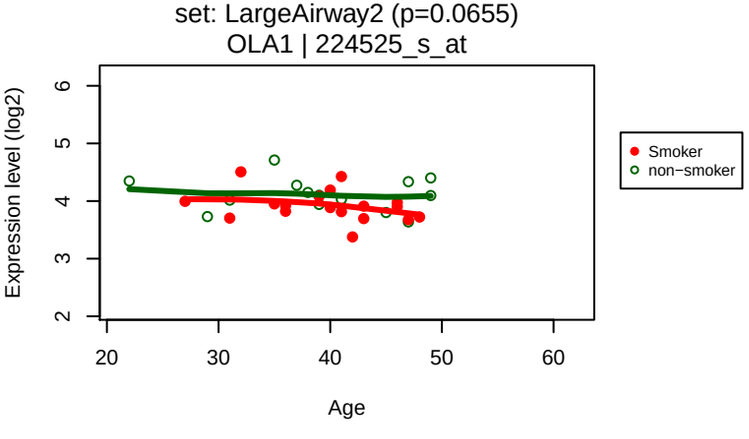

Gene expression ratio (smokers/nonsmokers)
gene: OLA1 | 224525_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2157
p=0.0407
p=0.0013
p=0.1153
p=0.0655
p=0.3393
p=0.4666
p=0.5225
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

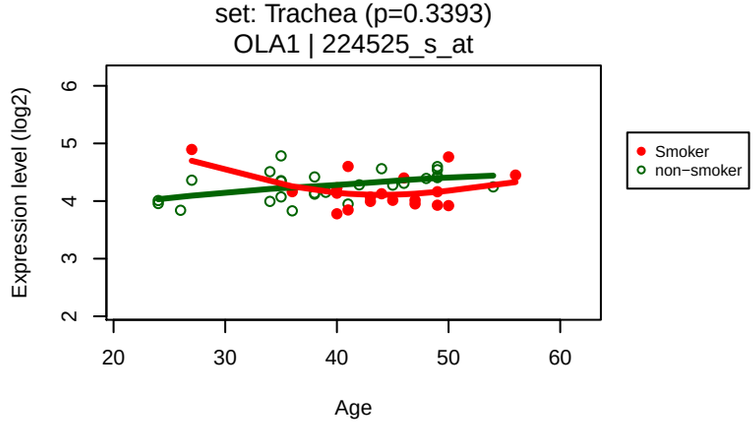

Gene expression ratio in Males
gene: OLA1 | 224525_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3565
p=0.0442
p=0.3968
p=0.3881
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: OLA1 | 224525_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0949
p=0.8699
p=0.0267
p=0.8659
SmallAirway2
LargeAirway2
Trachea
Macrophage2

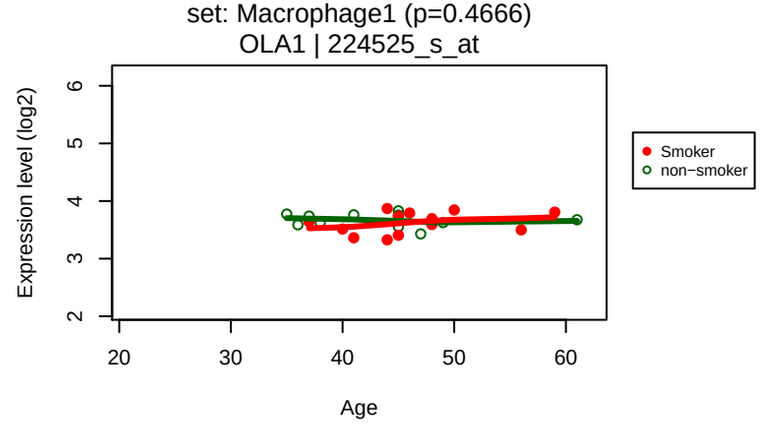

set: SmallAirway1 (p=0.2157)
OLA1 | 224525_s_at
Expression level (log2)
Age
Smoker
non−smoker

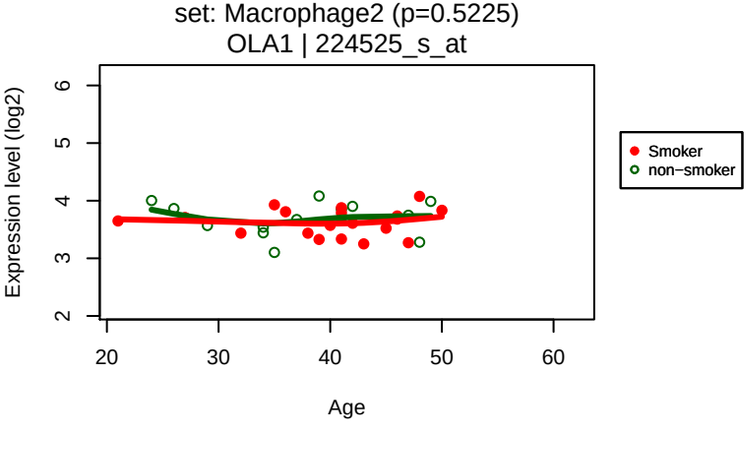

set: SmallAirway2 (p=0.0407)
OLA1 | 224525_s_at
Expression level (log2)
Age
Smoker
non−smoker

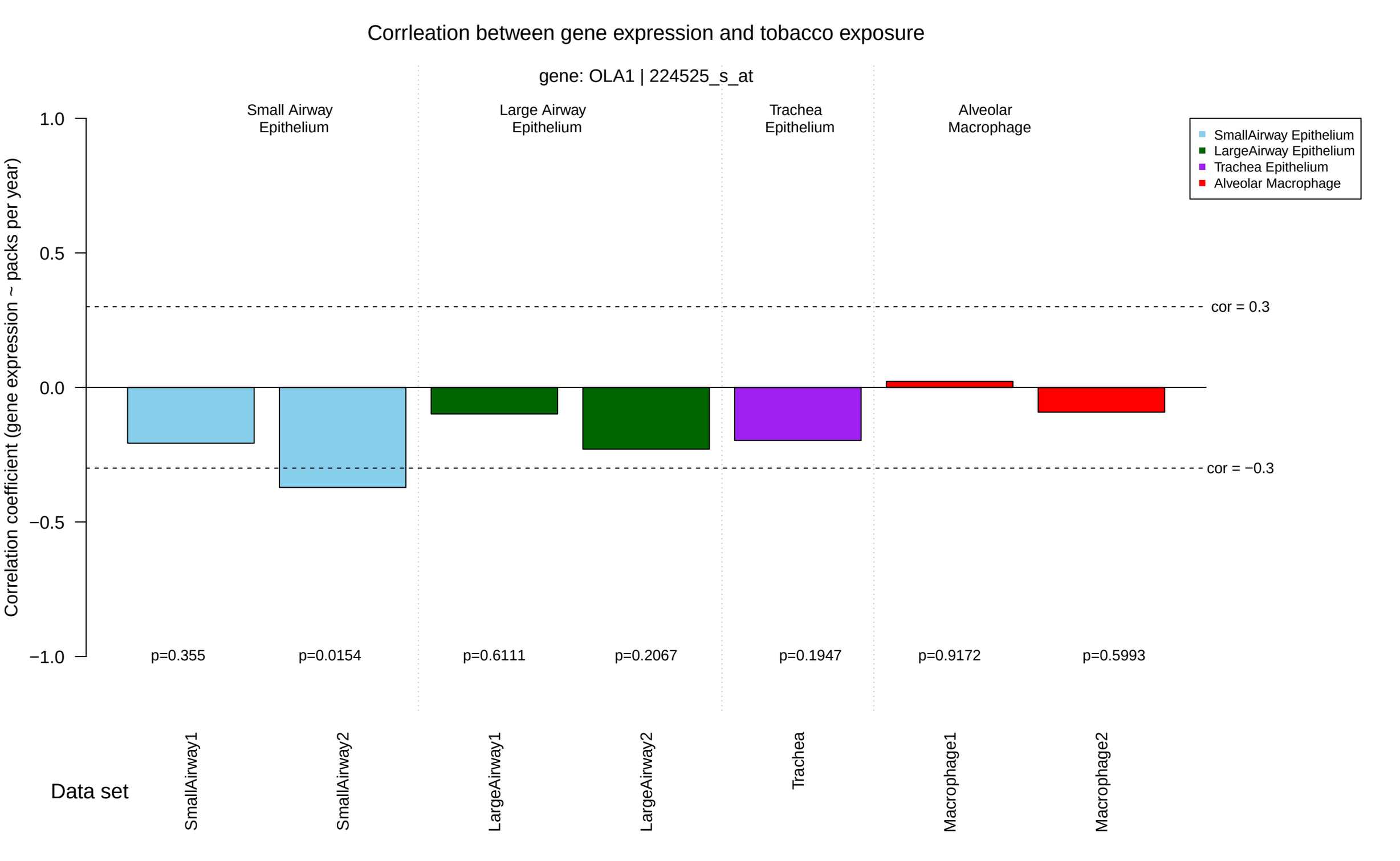

set: LargeAirway1 (p=0.1153)
OLA1 | 224525_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0655)
OLA1 | 224525_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3393)
OLA1 | 224525_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4666)
OLA1 | 224525_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5225)
OLA1 | 224525_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: OLA1 | 224525_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.355
p=0.0154
p=0.6111
p=0.2067
p=0.1947
p=0.9172
p=0.5993
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2